

# RAVI SONI

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## PROFILE: ELECTRONICS ENGINEER

- Seeks to understand the Bigger Picture. Asks 'why' and finds the answer.
- Dedicated electronics engineer with interests in embedded systems, control systems, robotics, and electronics hardware development. Highly proficient in Electro-mechanical system design, Vision guided robot design and forward kinematics robotic arm design. Highly skilled in C/C++, Python and Matlab.
- Avid learner. Curious to learn new concepts for the product development. Implement and generate new unique solutions that improve the performance and scalability of the product.
- Adept at working independently with little to no supervision or as part of a professional engineering team.
- Able to convey complex theories and engineering concepts in a unique way to a wide variety of audiences in a clear and accessible manner.

## EDUCATION

2013, Bachelor of Science, Electronics & Communication Engineering, Charotar University (AICTE approved), India.  
CGPA: 8.97/10.0

## TECHNICAL SKILLS

### IN THE MACHINE:

C Embedded Programming, Python, Shell Scripting, C/C++, Linux OS, Arduino & Processing, Raspberry Pi, Beaglebone Black, OpenCV library, MATLAB, Octave, Microsoft Office Suit, LabView, Multisim, PSPICE, Proteus, HFSS, Eagle, Keil, Code Composer Studio, ARM Cortex-M series programming, SketchUp 3D Modeling, ZigBee, Protocols: RS232, I2C, SPI, USB, CAN

### IN THE REAL WORLD:

Electronic Bench Test Equipment, Control System Design, Robotic Arm design, RFID & NFC Technology, Circuit Design & Debugging, Microcontroller/microprocessor (AVR, MSP430, STM32 Series), 3D Printing Technology, Signal Processing, Product Design, Project Management, Client Relations, Prototyping, Public Speaking, Large Group Leadership

## ENGINEERING EXPERIENCE

LIMBERLINK TECHNOLOGIES, BANGALORE, INDIA

July 2013-March 2015

JED-I Fellow | Hardware Engineer, Researcher, Instructor

- Facilitated JED-I (Joy Of Engineering, Design and Innovation) programs in engineering colleges and professional education in corporates helped me to improve my analytical, communication and technical skills.
- Trained engineering students by giving them Project-Based Learning (PBL). It is a unified engineering approach by combining Electronics, Mechanical, Computer Science and Aeronautical engineering concepts.
- Identified and redesigned hardware related control system problems in the product PYGMY, PORS & RAO Pvt. Ltd, Bangalore, India. Worked with STM32 processors, linear actuators, high speed motors and control system modeling.
- Worked on Kalman filter implementation in Self balancing robot which uses Inertial Measurement Unit (IMU). Designed Autonomous soccer robot using Computer Vision – Contour Tracking Algorithm and Raspberry Pi.
- Built multidisciplinary embedded projects like self-balancing robot, flying wing, MEMS based pedometer, navigation mobile robot and analog system design module- Oscillator, Filters, Amplifiers, PLL.
- Implemented and learnt CAN bus, STM32 programming, Raspberry Pi and OpenCV, Motion Planning, PID Control, Analog Circuit Design, PCB Manufacturing, Airfoil design, Wind-tunnel design, A\* and Dijkstra's algorithm, Artificial Intelligence in mobile robots, Scientific Computing.
- Worked on RFID based Access control System for the research laboratory called C-Camp located in Bangalore, India.

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ELECTRONICS & COMMUNICATION DEPARTMENT, CHAROTAR UNIVERSITY, INDIA

December 2012-May 2013

## Research Intern

- Worked on 'SPARSH': an artificial prosthetic arm under guidance of academic and industry researchers. It is an electromechanical project and prosthetic arm developed for amputees.
- Designed and manufactured different parts of arm like fingers, palm and elbow. Designed and fabricated challenging robotic hand having 15 Degrees-of-Freedom. Worked on ZigBee protocols to wirelessly control the robotic arm with our own arm and replicate our grip patterns.
- Electrodes and force sensors were fixed on the amputee's nerves to get the sensory data and AVR processors were used to process it.

iClub, CHAROTAR UNIVERSITY, INDIA

October 2011-June 2013

## Technical Committee Leader

- iClub (Innovative Club) is an In-house platform for students to develop research projects in iClub Lab and work closely with faculties and researchers from different departments, with different background and research skills. Directed and led several teams for the product development.
- Organized seminars on HFSS Antenna design and modeling, Design of an embedded systems, Computer networking. Delivered workshops on Scientific Computing, Embedded System design and Mobile robots.

## RESEARCH PROJECTS AND AWARDS

### PYGMY | LIMBERLINK TECHNOLOGIES, BANGALORE, INDIA

Identified and redesigned Hardware related Control System problems in this product. It is the Product of PORS & RAO, Bangalore. Enhanced the efficiency of the Hardware by working on PCB debugging, filters and PID control of the system.

### REAL TIME 3D SCANNER | LIMBERLINK TECHNOLOGIES, BANGALORE, INDIA

Designed 3D scanner, can scan the real objects using laser and cameras. Worked on image processing, implementation of triangulation method and actuators.

Winner: In-house Jed-I Project challenge, Bangalore.

### 'SPARSH': AN ARTIFICIAL PROSTHETIC ARM FOR AMPUTEES | CHAROTAR UNIVERSITY, INDIA

'SPARSH' is a Robotic arm for amputees, developed as final year research project. It is a muscle controlled prosthetic arm.

Nominee: Fab Replication Grant, FabLab Connect, Fab Foundation | April'15

Conference: Electronics Rocks'13, By Electronics For You (EFY) Magazine | Sept'13

Award: The Tally Innovation award, By Indian Institute Of Science (IISc) Bangalore | June'13

Business plan: Top10, Next Big Idea 2013, By Intel, DST of India & IIM Bangalore | May'13

### LOW POWER ELECTRONICS DIYA | TEXAS INSTRUMENTS INDIA, INDIA

Designed ultra-low power 'diya' (light) for Diwali using LEDs, MSP430 controller and electronics power circuit.

Winner: Low power electronics Diya contest, Texas Instruments, Bangalore.

### AUTONOMOUS MAZE ROBOTS | TECHFEST'12 IIT BOMBAY, INDIA

Designed mobile robots to solve maze, communicate and transport the blocks. Implemented A\* algorithm and path-planning.

## RESEARCH INTERESTS

ELECTRICAL: Control System Design, Embedded System Design, Medical & Service Robotics, Image & Video Signal Processing

MECHANICAL & AERONAUTICS: Product Design, Mechanisms, Airfoil & Wind Tunnel Design, Quadcopter & Wing Development

COMPUTER SCIENCE: Computer Vision, Algorithms, Augmented Reality, Nand2Tetris Computer Design

**ONLINE COURSES:** Hardware Security, *University of Maryland* | Image and Video Processing, *Duke University*